CHAPTER 17
REVIEW

KEY QUESTIONS AND TERMS

1. d DOK 1
2. a DOK 1
3. b DOK 1
4. A cell that has a large surface area to volume ratio would be more efficient because it has a larger cell membrane through which nutrients and wastes may pass. DOK 2
5. Asexual reproduction is simple and efficient. It allows an organism to produce many offspring quickly. Sexual reproduction is slower, but it introduces genetic variation in offspring, which can increase offspring survival rates. DOK 2
6. b DOK 1 Centromere
7. c DOK 2 12
8. d DOK 1 a cell plate
9. Student diagrams should show a cell that had *x* amount of DNA at the beginning of the S phase and show that after the S phase, the cell has 2*x* amount of DNA. Possible description: To prepare a eukaryotic cell's DNA for mitosis, the cell grows in the G_1 phase. In the S phase, new DNA is synthesized as the chromosomes are replicated. At the end of the S phase, the cell contains twice as much DNA as it did at the beginning. Possible analogy: A teacher assigns a project to partners. The teacher gives one copy of the assignment to each pair. The pair copies the assignment, doubling the number of instructions, and then they each student takes one copy home. DOK 3
10. It used to refer to a phase where it was thought there was little or no activity. Now we understand that a lot happens during that time, so it is now divided into three phases—G_1, S and G_2. DOK 3
11. prophase: DNA condenses, a spindle forms, and centrioles move to opposite poles; metaphase: centromeres of duplicated chromosomes line up across center of cell and spindle fibers attach centromeres to spindle; anaphase: sister chromatids separate and move to opposite ends of cell; telophase: chromosomes decondense, and a nuclear envelope forms around each group of chromosomes DOK 1
12. The phase is cytokinesis. In most animal cells, the cell membrane is drawn inward until the cytoplasm is pinched into two nearly equal parts. Each part contains its own nucleus and cytoplasmic organelles. In plants, a cell plate forms halfway between the divided nuclei. The cell plate gradually develops into cell membranes that separate the two daughter cells. A cell wall then forms in between the two new membranes. DOK 2
13. a DOK 1 They respond to events occuring inside cell
14. c DOK 1 cyclins
15. In the lab, cells grow and divide until they come into contact with each other. Once they make contact, they stop dividing. When a bone breaks, the cells are no longer in contact with each other, so the bone cells start dividing. Once the break is closed, the cells are in contact and stop dividing, and the wound is considered healed.
16. Cyclins are proteins that regulate the timing of the cell cycle. DOK 1
17. Both: undergo cell division; both have DNA; both have normal organelles Cancer cells: ability to spread; do not undergo normal apoptosis; lose the ability to regulate cell division; Noncancerous cells: undergo apoptosis; undergo normal cell division; DNA is properly duplicated during mitosis DOK 2
18. a DOK 1 totipotent or b. pluripotent
19. d DOK 1 They can grow into many tissue types
20. The inner cells of the blastocyst develop into the actual body of the embryo. The outer cells form tissues that will attach the embryo to its mother. DOK 2
21. It may be possible to reverse the damage caused by a severe heart attack by injecting stem cells into the damaged heart, where they differentiate into new heart muscle cells. DOK 3

CHAPTER 17
REVIEW

KEY QUESTIONS AND TERMS

17.1 Cell Growth, Division, and Reproduction

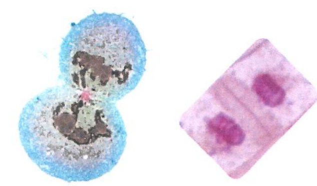
HS-LS1-4

1. The rate at which materials enter and leave the cell depends on the cell's
   a. volume to DNA ratio.
   b. weight to surface area ratio.
   c. DNA to surface area ratio.
   d. surface area to volume ratio.
2. During interphase, the cell must first
   a. duplicate its genetic information.
   b. decrease its volume.
   c. increase its number of chromosomes.
   d. decrease its number of organelles.
3. The process that increases genetic diversity within a population is
   a. asexual reproduction.
   b. sexual reproduction.
   c. budding.
   d. binary fission.
4. Explain which cell would be more efficient at obtaining nutrients and releasing wastes—a cell that has a large surface area to volume ratio, or a cell that has a small surface area to volume ratio.
5. Describe asexual and sexual reproduction as species survival strategies.

17.2 The Process of Cell Division

HS-LS1-4

6. Sister chromatids are attached to each other by a structure called the
   a. centriole.
   b. centromere.
   c. spindle.
   d. chromosome.
7. If a dividing cell initially has 12 chromosomes, how many chromosomes will each of its daughter cells have after mitosis and cytokinesis?
   a. 4 c. 12
   b. 6 d. 24
8. In plant cells, what forms during cytokinesis that does not form in animal cells?
   a. the nuclear membrane
   b. a centromere
   c. a cell membrane
   d. a cell plate
9. Use a diagram and words to show the steps involved in preparing a eukaryotic cell's DNA for mitosis. Use an analogy to explain why you think this helps the cell divide the DNA evenly.
10. The term *interphase* used to refer to an in-between phase in cell division. Why has this view changed, and what has changed in our understanding of this part of cell division?
11. List the following phases of mitosis in the correct sequence, and describe what happens during each phase: anaphase, metaphase, prophase, and telophase.
12. Identify the phase of mitosis that the animal and plant cells are in. Describe the events of this phase, and in particular identify any differences between the animal and plant cells during this phase.



Animal Cell Plant Cell

17.3 Regulating the Cell Cycle

13. Which statement does NOT describe external regulatory proteins?
   a. They respond to events occurring inside a cell.
   b. Growth factors are one group of external regulatory proteins.
   c. They can speed up or slow down the cell cycle.
   d. Some can cause cells to slow down or stop their cell cycles.
14. The timing of the cell cycle in eukaryotic cells is controlled by a group of closely related proteins known as
   a. chromatids.
   b. centromeres.
   c. cyclins.
   d. centrioles.
15. Cell division is regulated by both internal and external proteins. Explain how the observations of cultured cells can be used to understand how a broken bone heals.
16. Describe the role of cyclins.
17. Use a Venn diagram to differentiate between a cancerous and noncancerous cell.

17.4 Cell Differentiation

HS-LS1-4

18. Which type of cell has the potential to develop into the widest variety of differentiated cells?
    a. totipotent
    b. pluripotent
    c. multipotent
    d. differentiated
19. What property of stem cells makes them so attractive as a medical treatment?
    a. They can grow into heart tissue.
    b. They can be cultured into whole organs.
    c. They can easily be collected from reproductive tissue.
    d. They can grow into many different tissue types.
20. What is the difference between the inner and outer cells of the blastocyst?
21. Describe why stem cell injection therapy for heart attacks and other disorders is so attractive to medical research.
22. Describe the three methods for harvesting stem cells. For each method, explain the benefits of the method and the ethical concerns related to the method.

CRITICAL THINKING

HS-LS1-4

23. **Reason Quantitatively** Two objects modeling cellular shapes are shown. All of the sides in Shape A are equal in size.

Compare the ratio of surface area to volume for the two objects. Then use your results to explain how the long, flattened shapes of some organisms, such as single-celled paramecia, help them survive.

24. Consider the difference between the genetic variability of sexually and asexually reproducing organisms.
    a. **Make a Claim** Write a sentence that states whether sexually or asexually reproducing organisms would have an advantage in an environment that undergoes frequent changes.
    b. **Use Evidence** Use evidence from the chapter to defend your claim.
    c. **Use Reasoning** Connect your evidence to your claim. Explain why you chose to make the claim that you did.
25. **Compare and Contrast** Differentiate between DNA replication in prokaryotes and DNA replication in eukaryotes.
26. **Summarize** During mitosis, it is essential for the organism to pass on the correct number of copies of DNA. Use a chart to summarize what occurs to the DNA during interphase, prophase, metaphase, anaphase, and telophase to ensure that each daughter cell receives the correct amount of DNA.
27. **Synthesize Information** Cyclin is produced inside the cell and regulates the cell cycle. Suppose scientists claim that they have discovered a drug that can cure cancer by absorbing all of the cyclin in dividing cells. Determine how such a drug would affect cellular reproduction in an organism.
28. **Draw Conclusions** Scientists are investigating a chemical that prevents DNA synthesis.
    a. In which phase of mitosis does the chemical act?
    b. Why might scientists be interested in this as a treatment for cancer cells?
29. **Predict** A scientist is working on a new treatment for cancer that re-engineers the function of the gene p. 53 and turns it back on. What would be the effect of this treatment?
30. **SEP Construct an Explanation** A blastocyst contains two main types of cells: differentiated outer cells and totipotent inner cells. How do you think this design helps the blastocyst protect a growing embryo and ensure its proper development once it implants on the uterine wall?
31. **SEP Evaluate Models** The model in **Figure 17-18** on stem cell injection therapy appears to be a straightforward process of removing the bone marrow cells and injecting them directly into the damaged area. Why would this seem to be a natural solution, and what do you think are some of the reasons why this treatment has not become mainstream yet?

22. **(1)** Embryonic stem cells are collected from a blastocyst. They are desirable because they are pluripotent. Their collection results in the destruction of an embryo, which many people view as unethical because one life is destroyed for another, and the embryo does not have a choice in this type of collection. **(2)** Adult stems cells are collected from a small number of places in an adult, including bone marrow and hair follicles. They are less desirable because they are only multipotent. Adult stem cells are collected from consenting adults, so there are few ethical concerns. **(3)** Induced pluripotent stem cells are differentiated adult cells that have been reprogrammed to a pluripotent embryonic state. They are advantageous because they come from willing donors and appear to be pluripotent. These cells allow researchers to work on developing therapies using stems cells and to understand differentiation conditions without destroying embryos. **DOK 3**

REVIEW

CRITICAL THINKING

23. A: Surface area = $9 \times 9 \times 6 = 486$ mm^2;
Volume = $9 \times 9 \times 9 = 729$ mm^3;
SA/V = 486/729 = 0.67
B: Surface area = $(36 \times 6 \times 4) + (6 \times 6 \times 2)$
= 864 + 72 = 936 mm^2;
Volume = $36 \times 6 \times 6 = 1296$ mm^3;
SA/V 936/1296 = 0.72
The SA/V ratio of the long, flattened shape is greater. An organism with this sort of shape would be able to take in needed materials and expel wastes more efficiently than a cube-shaped organism. **DOK 3**
24. **a.** Sample answer: Organisms that reproduce sexually are better suited to environments that undergo frequent changes.
    b. Sample answer: Evidence that supports this claim includes the fact that organisms that reproduce sexually produce more genetic variability. This variability allows species to better adapt to environmental changes because it allows for new combinations of genes.
    c. Sample answer: Organisms that reproduce asexually produce offspring with the same genes. This lack of genetic variation can make survival more unlikely in an environment that changes rapidly. **DOK 3**
25. In a prokaryotic cell, the single chromosome replicates when the cell reaches a certain size. In a eukaryotic cell, the numerous chromosomes replicate in the S phase of interphase. **DOK 2**
26. Interphase: DNA is replicated. Prophase: DNA is condensing to sister chromosomes around a centromere. Metaphase: DNA is aligning along the midline of the cell as paired sister chromosomes; the spindle fibers from each pole attach to the centromere to ensure each receives a copy of the sister chromosome. Anaphase: Half of each sister chromosome is pulled toward the centriole at either pole. Telophase: The cell is divided into two daughters, the nuclear membrane reforms, and the DNA is unpacked. **DOK 2**
27. If scientists could eliminate the cyclin in a cell, theoretically, the cell would not divide. **DOK 3**
28. **a.** The chemical would act in the S phase.
    b. It would be a chemical that would prevent cancer cells from replicating successfully because the DNA would not replicate. **DOK 3**
29. It would enable the p53 protein to work properly again so that the cell cycle would be halted until all chromosomes are properly replicated. **DOK 3**
30. The outer cells protect the growing embryo from external signals that the cells might receive if they attached directly to the uterine wall. This is particularly important because the cells are able to develop into many cell types. **DOK 4**